KANGAROO-STYLE · GRADES 1-2 · FREE DIAGNOSTIC

Kangaroo-Style Grades 1-2 Diagnostic Test

24 questions	75 min time	No calculator

Not affiliated with or endorsed by Math Kangaroo.

1. 24 multiple-choice problems in three parts, easy to hard. Each has one right answer: write its letter (A to E) in the box. No calculator.
2. Part 1 problems are worth 3 points, Part 2 are worth 4 and Part 3 are worth 5 (96 points in all). A blank box scores zero, so make your best try.
3. Stuck? Skip it and come back. After the test, scan the code next to any problem you missed to watch the key move for it.

Name _____ Date _____ Score _____ / 96

PART 1 · 3 POINTS EACH

Problems 1 to 8

1 Leo has 16 marbles. He puts 9 of them in a bag. How many of his marbles are not in the bag?

(A) 3 **(B)** 4 **(C)** 5 **(D)** 6 **(E)** 7

Stuck? Scan for the key

Answer ☐

2 Mae gives this picture a half turn, so it is upside down. Which picture does she get?

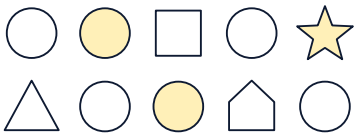


(A)  **(B)**  **(C)**  **(D)**  **(E)** 

Stuck? Scan for the key

Answer ☐

3 Some shapes are white and some are yellow. How many of the circles are white?

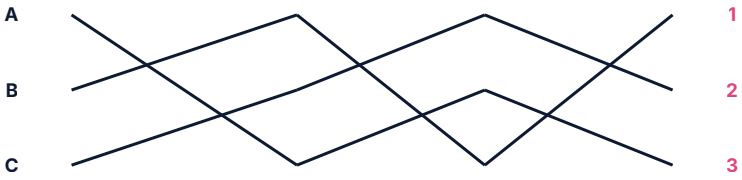


(A) 0 **(B)** 1 **(C)** 2 **(D)** 3 **(E)** 4

Stuck? Scan for the key

Answer ☐

4 Three strings cross each other. Follow the string that starts at B. Which number is at its other end?



(A) 1 **(B)** 2 **(C)** 3 **(D)** 4 **(E)** none

Stuck? Scan for the key

Answer ☐

5 Which number is 3 more than 8 and also 5 less than 16?

(A) 5 **(B)** 10 **(C)** 11 **(D)** 13 **(E)** 21

Stuck? Scan for the key

Answer ☐

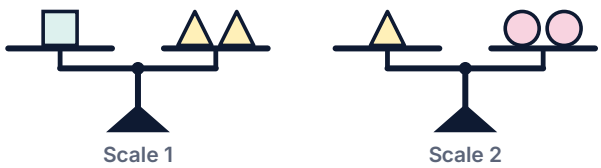
6 Mo plants 4 bushes in a row. From each bush to the next one is 3 m. How far is the first bush from the last one?

(A) 9 m **(B)** 10 m **(C)** 11 m **(D)** 12 m **(E)** 15 m

Stuck? Scan for the key

Answer ☐

7 The scales are level. A square has the same weight as how many circles?



(A) 3 **(B)** 4 **(C)** 5 **(D)** 6 **(E)** 7

Stuck? Scan for the key

Answer ☐

8 Pat skip-counts by 3 up to 9, then starts over: 3, 6, 9, 3, 6, 9, and so on. Which number does Pat say 11th?

(A) 3 **(B)** 5 **(C)** 6 **(D)** 9 **(E)** 12

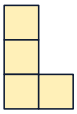
Stuck? Scan for the key

Answer ☐

PART 2 · 4 POINTS EACH Problems 9 to 16

9 A piece is missing from this dark board. Which piece fits into the white hole exactly? You may turn it.



(A)  **(B)**  **(C)**  **(D)**  **(E)** 

Stuck? Scan for the key

Answer ☐

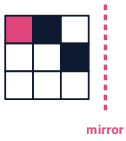
10 Lina rolls two dice and gets 8 dots in all. Which pairs of numbers could the dice show? Count how many different pairs there are (2 and 6 is the same pair as 6 and 2).

(A) 0 **(B)** 1 **(C)** 2 **(D)** 3 **(E)** 4

Stuck? Scan for the key

Answer ☐

11 Dot holds her card up to a mirror, as in the picture. Which picture does the mirror show?

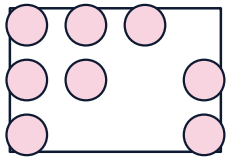


(A)  **(B)**  **(C)**  **(D)**  **(E)** 

Stuck? Scan for the key

Answer ☐

12 This box should hold 3 rows of 4 cupcakes. Some cupcakes have been eaten. How many are missing?



(A) 2 **(B)** 3 **(C)** 4 **(D)** 5 **(E)** 8

Stuck? Scan for the key

Answer ☐

13 A class has 12 girls and 10 boys. Today 5 of the girls are at home. How many girls are in class today?

(A) 5 **(B)** 7 **(C)** 15 **(D)** 17 **(E)** 22

Stuck? Scan for the key

Answer ☐

14 A tap fills 6 cups of water in 2 minutes. How many cups does it fill in 5 minutes?

(A) 10 **(B)** 12 **(C)** 13 **(D)** 15 **(E)** 30

Stuck? Scan for the key

Answer ☐

15 This rectangle is 7 cm long and 3 cm wide. How long is the edge all the way around it?



(A) 10 cm **(B)** 14 cm **(C)** 17 cm **(D)** 20 cm **(E)** 21 cm

Stuck? Scan for the key

Answer ☐

16 Every row and every column of this board needs the numbers 1, 2 and 3. Some are written in already. What number belongs where the question mark is?

1		
	3	
?		2

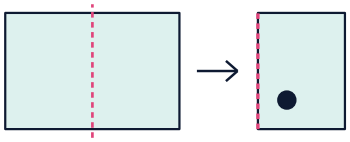
(A) 1 **(B)** 2 **(C)** 3 **(D)** 4 **(E)** There is no way to tell

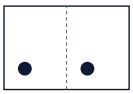
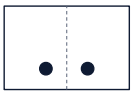
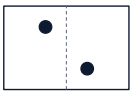
Stuck?
Scan for the key

Answer ☐

PART 3 · 5 POINTS EACH Problems 17 to 24

17 Eva folds her paper along the dashed line, left half onto right half. She punches one hole. Which picture shows the paper opened up?



(A)  **(B)**  **(C)**  **(D)**  **(E)** 

Stuck?
Scan for the key

Answer ☐

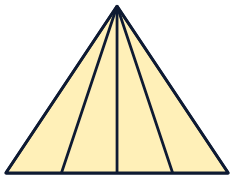
18 Kim, Lou and Max each have a bike: one red, one blue and one green. Kim's bike is not red. Lou's bike is blue. What color is Max's bike?

(A) red **(B)** blue **(C)** green **(D)** yellow **(E)** There is no way to tell

Stuck?
Scan for the key

Answer ☐

19 Ken's big triangle is cut by three straight lines that all start at the top corner. Count every triangle, big and small. How many are there?



(A) 6 **(B)** 8 **(C)** 10 **(D)** 11 **(E)** 12

Answer ☐

Stuck? Scan for the key

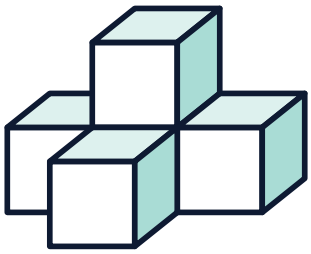
20 Mia and Zoe have 14 shells together. Mia has 8 of them. Zoe and Ivy have 10 shells together. How many shells does Ivy have?

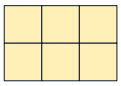
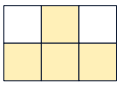
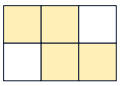
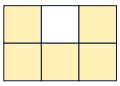
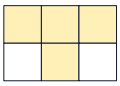
(A) 0 **(B)** 1 **(C)** 2 **(D)** 3 **(E)** 4

Answer ☐

Stuck? Scan for the key

21 Looking down from the ceiling at this cube building, what shape does Ava see?



(A)  **(B)**  **(C)**  **(D)**  **(E)** 

Answer ☐

Stuck? Scan for the key

22 A bus left school with some kids on it. At the first stop, 4 kids got off. At the next stop, 3 kids got on. Now 12 kids are on the bus. How many were on it when it left school?

(A) 5 **(B)** 10 **(C)** 11 **(D)** 12 **(E)** 13

Answer ☐

Stuck? Scan for the key

23 Some cats and some birds are in a yard. Ivan counts 5 heads and 14 legs. How many of the animals are cats?

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4 **(E)** 5

Stuck?
Scan for the key

Answer ☐

24 Sid's sock drawer holds 5 red socks and 5 blue socks. It is dark, so he cannot see the colors. What is the fewest socks he must pull out to be sure of a matching pair?

(A) 2 **(B)** 3 **(C)** 4 **(D)** 6 **(E)** 10

Stuck?
Scan for the key

Answer ☐

End of test. Check that every box has an answer.

KANGAROO-STYLE · GRADES 1-2 · ANSWER KEY

Diagnostic Test: Answers

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Each problem uses one tool from the channel. For any you missed, scan the code next to that problem on the test to watch the lesson, then try the problem again tomorrow without looking. Score: 3 points for each right answer in 1 to 8, 4 points in 9 to 16, 5 points in 17 to 24 (96 in all).

1 **E 7** Tool #78: Whole Minus Part
Whole minus part: 16 marbles in all, 9 in the bag, so 16 − 9 = 7 are not.

2 **A** Tool #65: Turn It in Your Head
A half turn sends each corner to the opposite corner: the dark square at the top left goes to the bottom right, and the pink one goes to the top left.

3 **E 4** Tool #79: Test Before You Count
Test each shape twice: is it a circle? Is it white? Two in each row pass both tests: 4.

4 **A 1** Tool #75: Trace the Line
Put your finger on B and slide along the string to the end. At each crossing, stay on the same string.

5 **C 11** Tool #73: Check Each Choice
Check each choice against both clues: 8 + 3 = 11 and 16 − 5 = 11. Only 11 passes both.

6 **A 9 m** Tool #66: Draw the Picture
Draw it: 4 bushes have 3 gaps between them. 3 + 3 + 3 = 9 m.

7 **B 4** Tool #74: Balance the Scales
Scale 1: a square weighs 2 triangles. Scale 2: a triangle weighs 2 circles. So a square weighs 2 + 2 = 4 circles.

8 **C 6** Tool #5: Find the Cycle
The pattern 3, 6, 9 has 3 numbers. 11 = 3 + 3 + 3 + 2, so number 11 is like number 2: it is 6.

9 **B** Tool #69: Count Then Fit
Count, then fit: the hole has 3 squares in a corner shape. Only one piece has 3 squares in that shape.

10 **D 3** Tool #67: Make a Neat List
Make a neat list, smaller number first: 2 and 6, 3 and 5, 4 and 4. A die has no 7, so 1 and 7 is out. That is 3 pairs.

11 **D** Tool #70: Mirror Swap
A mirror swaps left and right but keeps top and bottom. The pink and dark squares in the top row move to the right side.

12 **C 4** Tool #76: Fill In and Count
Fill in the full box: 3 rows of 4 make 12 cupcakes. 8 are there, so 12 − 8 = 4 are missing.

13 **B 7** Tool #81: Circle the Ask
Circle the ask: GIRLS in class. 12 − 5 = 7. The boys do not matter.

14 **D 15** Tool #10: One Unit First
One unit first: in 1 minute it fills 6 ÷ 2 = 3 cups. In 5 minutes: 5 × 3 = 15 cups.

15 **D 20 cm** Tool #72: Label Every Side
Label every side: the sides across are 7 and 7, the sides up and down are 3 and 3. 7 + 7 + 3 + 3 = 20 cm.

16 **C 3** Tool #80: Pickiest Box First
Pickiest box first: the ? square's column already has a 1 and its row already has a 2, so it must be 3.

17 **D** Tool #71: Unfold Step by Step
The hole goes through both layers, so it gets a twin on the other side of the fold, just as far from it and at the same height.

18 **A red** Tool #68: Cross Out the No's
Cross out the no's: Lou has blue. Kim is not red, so Kim has green. Red is left for Max.

19 **C 10** Tool #26: Count by Size
Count by size: 4 small triangles, 3 made of two small ones, 2 made of three and 1 made of all four. 4 + 3 + 2 + 1 = 10.

20 **E 4** Tool #78: Whole Minus Part
Whole minus part twice: Zoe has 14 − 8 = 6. Then Ivy has 10 − 6 = 4.

21 **E** Tool #65: Turn It in Your Head
From above you see the top of every column, tall or short: the whole back row and just the middle of the front row.

22 **E 13** Tool #39: Work Backwards
Work backwards: before the 3 got on there were 12 − 3 = 9. Before the 4 got off there were 9 + 4 = 13.

23 **B 2** Tool #11: Pretend All the Same
5 heads means 5 animals. Pretend all 5 are birds: 10 legs. Each cat adds 2 more legs. 14 − 10 = 4 = 2 × 2, so there are 2 cats.

24 **B 3** Tool #14: Worst Case
Worst case: the first two socks are different colors. The third sock must match one of them. So 3.